

What's the best way to learn a new language?

Krupa Padhy uncovers how we really learn foreign languages – in a dual challenge involving both Portuguese and Mandarin.

There was a time when my oversized hardback Collins Roberts French dictionary took pride of place on my bookshelf of my student accommodation. I owned an edition from the late 1980s, almost 1,000 pages long, handed down from my elder brothers. It travelled with me to Paris in the early 2000s, taking up half the space of my little case as a non-negotiable.

It was a sad day when a decade later, bursting at the seams of our one-bed flat with two babies, I decided it had to go. It had gathered dust since leaving university but had equally screamed that I had once been serious about language-learning.

Multilingualism has always been a part of my fabric. I was born into a Gujarati-speaking household, my Indian-origin parents having immigrated to the UK from Tanzania in the 1970s. My reading and writing skills were topped up with lessons at the local temple every Saturday as a kid. In 1995, Zee TV arrived in the UK on cable network, and I became hooked on watching cheesy Hindi serials every evening with the subtitles on. I took French to degree level and headed for my year abroad to Paris. Finally, a tinge of Spanish came to me after a few terms of evening classes. All these languages (bar the holiday-Spanish) have taken time and commitment.

Understandably maybe, I've reacted reluctantly to the countless advertisements on my Instagram feed promising to teach me a language in 30 days (if not sooner) by giving up less than 30 minutes a day.

term [brain health](#) and happiness are well noted, so no regrets there. But had my four years of studying a language to degree level conjugating verbs and memorising vocabulary become an outdated way of learning?

Along with the promise of becoming fluent at lightning speed, a range of new methods and [technologies](#) have [transformed](#) how we pick up languages in an increasingly [time-poor](#) age. One is "microlearning", [an approach](#) that breaks down new information into small chunks that are meant to be absorbed quickly, sometimes within minutes or even seconds. It's rooted in a concept known as the [forgetting curve](#), which states that when people take in large amounts of information, they remember less of it over time.

In addition, there's a wealth of new technologies, from [chatbots](#) offering instant feedback, [to virtual reality and augmented reality](#)

native speakers. However, [some argue](#) that the promise of fast fluency misses crucial elements of actually learning to speak to people in another language, such as developing cultural understanding and nuance.

So, with all this choice, what's actually the best, science-backed way to learn a language? To find out, I teamed up with two researchers at Lancaster University's Language Learning Lab: Patrick Rebuschat, a professor of linguistics and cognitive science, and Padraic Monaghan, a professor of cognition in the department of psychology. They let me try out an experiment they designed to mirror language-learning in the real world, and reveal how our brain picks up and makes sense of new words and sounds. The tasks basically simulate how we would cope if we were dropped into a foreign country with an unknown language, and just had to use our innate skills to figure out the new, mysterious

make sense of them.

Having not learnt a language in two decades, I was about to learn some Mandarin and Portuguese. Over six days I would be spending just 30 minutes per day on the tasks and tests. I was to complete them, not ask any questions and wait until the end of the experiment for feedback.

Monaghan explains that such experimental studies are used to establish how people begin to get a foothold in a language.

I was intentionally not told from the outset what the tasks were about. But the researchers later explained that they were designed to activate my brain's cross-situational learning (CSL) skills: that's our natural, instinctive ability to use statistics to gradually work out the meanings of words and basic grammar. It is essentially our brain's inherent ability to recognise patterns and

well with each other) based on the frequency of their use.

"People can learn very, very fast simply by keeping track of the statistics in the environment," says Rebuschat. "This type of task is designed to mimic real-world learning under immersion settings, where things are often ambiguous and we rarely receive immediate feedback."

Ahead of starting the experiment, I assumed that with my prior knowledge of French and basic Spanish, Portuguese would come naturally. Mandarin on the other hand was for me as foreign as a foreign language gets.

I'd also predicted that as I had done with most of my other languages, lesson one would comprise of basic greetings. Far from it.

"If you were dropped into Portugal, Brazil, or another Portuguese-speaking country, the language you encounter would not unfold in a

explains Rebuschat. "Instead, you would hear a wide range of language in context: people ordering food in cafés, conversations on the street, a football commentary in the background."

Thus, my exercise with Portuguese was to choose whether the word or sentence I was hearing matched one of two scenes, both featuring animated animals. This continued on repeat across three days, an example of statistical learning in action, says Rebuschat. "It is a basic learning ability that humans use from [infancy](#)

– before infants know any language at all – to pick up patterns in the world around them. We use it to learn regularities in sounds, images, and events over time."

I was quick to lean on my prior language knowledge. I know for example in Hindi *saap* means snake, and upon hearing the word *sapo*

Soon after, I figured out that each noun appeared in both singular and plural forms performing one of four physical actions like pushing or pulling. The grammar was somewhat trickier but not unfamiliar from the French I had studied.

By day three of Portuguese, results showed my accuracy sat consistently between 90–100%, which I was told was higher than the typical English-speaking learner (presumably, because I was able to use those insights from my other languages). My brain was extracting meaning based on the frequency upon which the same nouns and verbs were appearing on screen.

My Mandarin learning journey started out somewhat differently.

As with Portuguese, I completed four short tasks and tests each day, but this time I was matching 12 incomprehensible sounds to images of 12 never-seen-before objects. As I

saying out loud were in fact Mandarin tones, which are a core feature of the language as a different tone can change the meaning of a word. Each made-up word was assigned to a specific object. Using artificial words, known as pseudowords, allows researchers to compare results and improvements fairly because students can't draw on prior knowledge.

At times, repeating the same tones made me comatose and admittedly, I came to my answers with zero scientific reasoning. *Lu-fah* for example sounded like a loofah which I matched with an object that had soft spikes!

Linguistics students who are native speakers of Mandarin at Lancaster University looked at how I did. By the end of my first session matching the pseudoword to the right made-up object I had reached 75% accuracy, rising to 80% in sessions two and three.

My production test results (where I was asked to

say the tone out aloud) were not as impressive, ranging from 38% rising to 55% by the third day, although I was reassured by Rebuschat that my scores were far above chance.

